

Student Guide – APT Project Definition

Phase 1 | Capstone Course (Degree Portfolio)

Duoc UC – Viña del Mar Campus

A. PART I

1. Personal Information

The table below presents the completed student and academic information requested:

Student Names	Bastian Vega, Santiago Vega, Lukas Martinez
National ID (RUT)	21.758.557-0, 21.128.181-2, 20.725.137-2
Degree Program	Computer Engineering (<i>Ingeniería en Informática</i>)
Campus	Viña del Mar

2. APT Project Description

The following section outlines the project name, applicable graduate competencies, and associated professional performance areas.

Project Name	Sumatoria+co
Performance Area(s)	- Software and Mobile Application Development - Artificial Intelligence Implementation - User Experience and Interface Design (UX/UI)
Graduate Competencies	- Deliver computing solution proposals through comprehensive process analysis in accordance with organizational requirements. - Manage IT and software projects, offering decision-making alternatives aligned with organizational requirements. - Build the architectural model of a systematic solution that effectively supports core business processes.

3. APT Project Rationale and Justification

Detailed breakdown of project relevance, scope, graduate profile alignment, professional interests, and development feasibility.

Relevance of the APT Project	The architecture and construction industry in Chile faces critical information fragmentation across project lifecycles. Currently, design, site operations, and administrative teams operate with isolated tools: Excel spreadsheets for budgeting, email for site meeting minutes, Dropbox for document storage, Monday.com for task tracking, and in-person meetings for follow-ups. This dispersion causes: • Wasted time locating up-to-date documentation. • Budget and schedule variances that remain undetected in timely windows. • Lack of end-to-end traceability regarding field decisions and operational revisions. • Heightened error risk resulting from outdated records across distributed teams. <i>Sumatoria+Co.</i>, as a strategic design and architectural project management firm in Chile, directly reflects this issue. While it uses market tools (Monday.com, Dropbox), these systems are neither integrated nor adapted to its specific workflow, which demands a unified overview from workshop design to final site delivery.
APT Project Description	Develop an MVP (Minimum Viable Product) of an Operational Construction Project Manager for <i>Sumatoria+Co.</i>, enabling field and central office teams to record, query, and track site meeting minutes and construction variances in real time, improving traceability and decision-making during the active building stage.
Alignment with the Graduate Profile	The Operational Manager MVP aligns directly with the DUOC UC Computer Engineering graduate profile: • Software Architecture & Quality: Designing a modular Single-Page Application (SPA) following structured design patterns and responsive UI standards. • Data Management: Relational data modeling for minutes, projects, and documents, coupled with a SQL query explorer for operational analytics. • Requirements Engineering: Eliciting operational needs from Construction Site Managers and Technical Inspectors (ITO), translating them into functional system modules. • Project Governance: An automated variance dashboard with early warning indicators and operational KPIs. • Security & Modularity: Implementing Role-Based Access Control (RBAC) and clean architectural separation to ensure long-term maintainability. • Mobile Development: Essential because field inspectors register minutes on-site via mobile devices, making the Tailwind CSS responsive interface indispensable. Together, these competencies ensure a viable technical solution that centralizes information, improves decision-making, and enhances operational efficiency in Chilean construction.
Alignment with Professional Interests	The project allows the team to apply cumulative academic engineering knowledge to an authentic enterprise setting. Leading the development lifecycle end-to-end for <i>Sumatoria+Co.</i> reinforces: • End-to-end system lifecycle visibility. • Technical leadership and project governance capabilities. • Architectural autonomy across backend and frontend stacks. • Creation of a production-grade portfolio project with measurable business value.
Project Feasibility	Timeline & Dedication: Structured across the 18 semester weeks, combining scheduled academic hours with continuous autonomous team effort. Technical & Material Resources: Each team member operates dedicated hardware. Version control and collaboration are centralized via GitHub. A dedicated financial budget is allocated for cloud hosting and operational services. External Enabling Factors: Strong bidirectional collaboration and open communication with <i>Sumatoria+Co.</i> leadership, alongside ongoing technical guidance from faculty advisors. Risks & Mitigation: Initial risks included feedback delays and scope expansion requests. The team mitigates this by prioritizing an MVP core for official academic grading; out-of-scope requests will be addressed in post-academic collaboration.

B. PART II

4. Objectives

General Objective	Develop and implement an MVP (Minimum Viable Product) of an Operational Project Manager for <i>Sumatoria+Co.</i> that centralizes the tracking of schedules, budgets, field inspection minutes, and construction variances, optimizing traceability and decision-making within the established academic timeframe.
Specific Objectives	1. Elicit and refine functional and non-functional system requirements through initial interactive mockups and structured feedback sessions with <i>Sumatoria+Co.</i> stakeholders. 2. Design the software architecture, relational data model, and UX/UI interface, ensuring cross-platform responsive ergonomics for both mobile field operations and central office desktops. 3. Build core functional modules covering project tracking, operational phases, schedules, budget management, render visualization, KPI monitoring dashboards, and version-controlled inspection minutes. 4. Validate system functionality, usability, and runtime performance through targeted technical testing suites and end-user verification sessions with designated key users.

5. Methodology

Methodological Framework:

The project follows an iterative and incremental Agile methodology, based on periodic functional deliveries and continuous feedback loops with *Sumatoria+Co.* This facilitates rapid requirements adaptation, managed functional scope, and early mitigation of deviations.

Work Phases:

1. **Discovery and Early Prototyping Phase:** Elicitation of operational requirements and construction of interactive functional mockups to validate baseline workflows.
2. **Refinement and Design Phase:** Functional specification tuning, software architecture design, relational database modeling, and detailed UX/UI layout definitions.
3. **Iterative Development Phase (Sprints):** Modular engineering of system features (projects, phases, budgets, minutes, schedules, and KPIs) backed by continuous integration and code reviews.
4. **Validation and Testing Phase:** Functional test suite execution and usability validation with end-users designated by *Sumatoria+Co.* to certify product viability.

Team Leadership and Roles:

- **Santiago Vega:** Project Manager and Primary Strategic Liaison with *Sumatoria+Co.* Oversees requirements validation, milestone agreements, and timeline governance.
- **Bastián Vega:** UX/UI and Frontend Lead. Directs interface design, design system implementation, and responsive client-side development for field and office use.
- **Lukas Martínez:** Backend and Data Architect. Leads relational schema modeling, server-side logic, API development, system integrations, and budget module structuring.

6. Project Evidences

Type	Deliverable Name	Description	Justification
Progress	Elicitation Records and Meeting Minutes	Video recordings and structured minutes from discovery and feedback sessions with <i>Sumatoria+Co.</i> leadership.	Certifies the formal requirements engineering process with an authentic client, demonstrating decision traceability.
Progress	Interactive Prototype and Budget Structuring Doc	Navigable interactive prototype alongside technical definitions and data structure of the project cost and budget module.	Evidences early UX flow validation, architectural feasibility, and core domain modeling prior to intensive coding.
Final	Centralized Source Code Repository (GitHub)	Complete source code repository for the MVP (Frontend, Backend, and DB migration scripts) with commit history and tags.	Certifies authorship, team collaboration, software quality, and application of architectural web standards.
Final	Technical Report and User Validation Results	Comprehensive report consolidating system architecture diagrams, user manuals, and results from key user testing sessions.	Formally demonstrates objective achievement, real-world operational suitability, and academic project completion.

7. Work Plan

Operational activity breakdown across the 18 academic semester weeks:

Competency	Activity Name	Description	Resources	Duration	Lead	Notes
Deliver computing solutions via process analysis	Act. 1: Requirements elicitation & refinement	Conduct work sessions with client to map processes, identify pain points, and record specifications.	PCs, video conference tools, meeting log templates.	Weeks 1–3	Santiago Vega	Enabler: High client openness. Risk: Frequent scope adjustments.
Build architectural models supporting business	Act. 2: UX/UI prototyping & architecture design	Design navigable prototype, responsive views, relational DB schema, and backend skeleton.	PCs, prototyping software (Figma/HTML), GitHub.	Weeks 3–5	Bastían Vega, Lukas Martínez	Enabler: Early client feedback on wireframes.
Manage IT projects and decision alternatives	Act. 3: Budget structuring & progress deliverable	Define project budget structures and assemble documentation for the Phase 1 progress report.	Spreadsheets, project docs, development IDEs.	Weeks 4–6	Lukas Martínez, Santiago Vega	Critical milestone closing Phase 1.
Build architectural models supporting business	Act. 4: Backend and Database development	Implement server logic, REST APIs, RBAC authentication, and persistence for projects, minutes, and budgets.	Local/cloud servers, SQL DB engine, Node.js, GitHub.	Weeks 6–12	Lukas Martínez (w/ Bastián, Santiago)	Enabler: Centralized repo. Risk: Concurrent module integration.
Deliver computing solutions via process analysis	Act. 5: Frontend & MVP module construction	Program responsive web/mobile interfaces: project views, schedules, render visualizer, and KPI panels.	IDEs (VS Code), Tailwind CSS, SPA frameworks.	Weeks 7–13	Bastían Vega (w/ Santiago, Lukas)	Guarantees field usability for on-site construction roles.
Manage IT projects and decision alternatives	Act. 6: System integration & staging deployment	Integrate frontend and backend layers, set up staging hosting, and seed mock datasets respecting confidentiality.	Cloud infrastructure, CI/CD pipelines, GitHub.	Weeks 13–14	Full Team (Bastián, Santiago, Lukas)	Involves planned infrastructure hosting investment.
Deliver computing solutions via process analysis	Act. 7: QA testing & client validation	Execute technical test suites, usability reviews, and formal validation sessions with client users.	Staging environment, feedback forms, mobile/desktop devices.	Weeks 15–16	Santiago Vega, Bastián Vega	Confirms compliance with agreed system objectives.
Manage IT projects and decision alternatives	Act. 8: Final report & academic closing	Author final technical report, assemble user guides, and organize final code repository and presentation.	Word processor, presentation tools, GitHub repo.	Weeks 17–18	Full Team (Bastián, Santiago, Lukas)	Final graduation evaluation gate for Capstone course.

8. Gantt Chart

Schedule distribution covering Phase 1, Phase 2, and Phase 3 across the 18 academic semester weeks:

Activity	Phase 1			Phase 2														
	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17	W18
1. Requirements with <i>Sumatoria+Co</i>	X	X	X															
2. UX/UI & Architecture Prototyping			X	X	X													
3. Budget Structuring & Phase 1 Report				X	X	X												
4. Backend & Database Development						X	X	X	X	X	X	X						
5. Frontend & MVP Module Construction							X	X	X	X	X	X	X					
6. Staging Integration & Deployment													X	X				
7. User Validation Testing (<i>Sumatoria+Co</i>)																X	X	
8. Final Report Consolidation & Defense																		